

Karakterisasi Material:

Transmissions Electron Microscopy (TEM)

Eka Nurfani

Program Studi Teknik Material, Institut Teknologi Sumatera

Introduction

ELECTRON MICROSCOPY:

- Scanning electron microscopy (SEM)
- **Transmission electron microscopy (TEM)**
- High voltage electron microscopy (HVEM)
- Scanning transmission electron microscopy (STEM)

Why should we use electron?

- TEM merupakan salah satu teknik pencitraan dengan resolusi spasial tertinggi
- Elektron memiliki kopling yang kuat dengan bahan dengan $\lambda_{\text{elektron}} < \lambda_{\text{sinar-X}}$
- Mampu mempelajari material dalam skala atomik

Why should we use electron?

Wave length of light and electron		
LIGHT	Species of light	Wave length (λ , nm)
	Electric wave	$10^6 \sim 10^{12}$
	Ultra red light	$10^3 \sim 10^5$
	Visible light	390~760
	Ultra violet light	13~390
ELECTROMAGNETIC WAVE	X-ray	0.05~10
	γ -ray	0.005~0.1
The wave length of ELECTRON is dependent on the accelerating voltage	Accelerating voltage	Wave length of electron (λ , nm)
	20 kV	0.00859
	120 kV	0.00335
	200 kV	0.00251
	1000 kV	0.00087

Why electrons?

□ Resolution of Electron microscope: ?



- Wave Behaviors
 - images and diffraction patterns
 - wavelength can be tuned by energies
- Charged Particle Behaviors
 - strong electron-specimen interactions
 - chemical analysis is possible

Light

- $p = h/\lambda$ (matter also)
- $p = E/c$
- $E = hf = hc/\lambda$

Matter

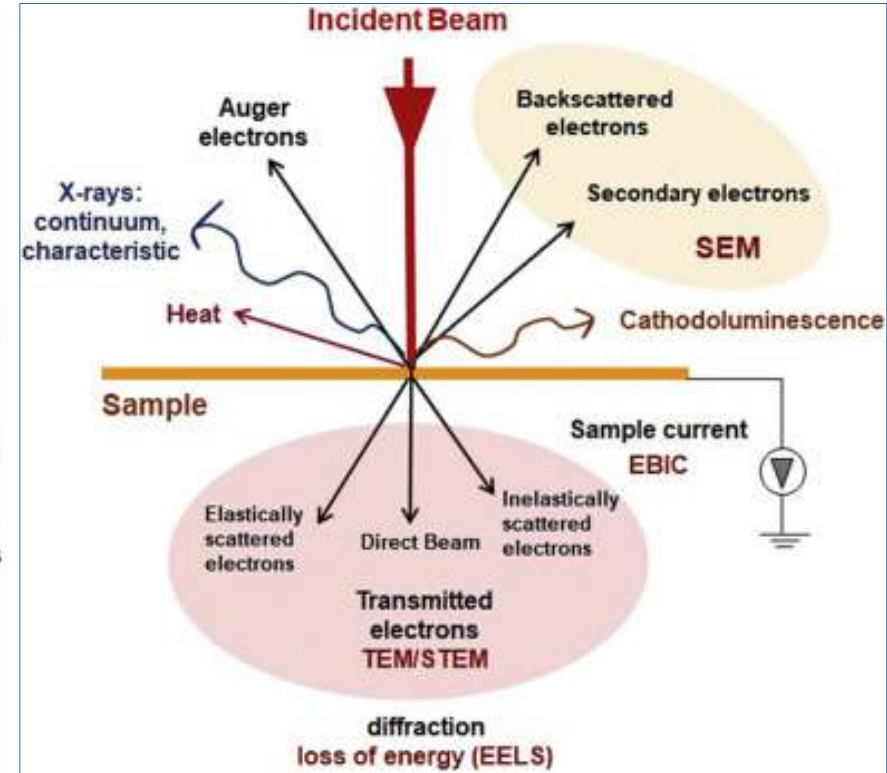
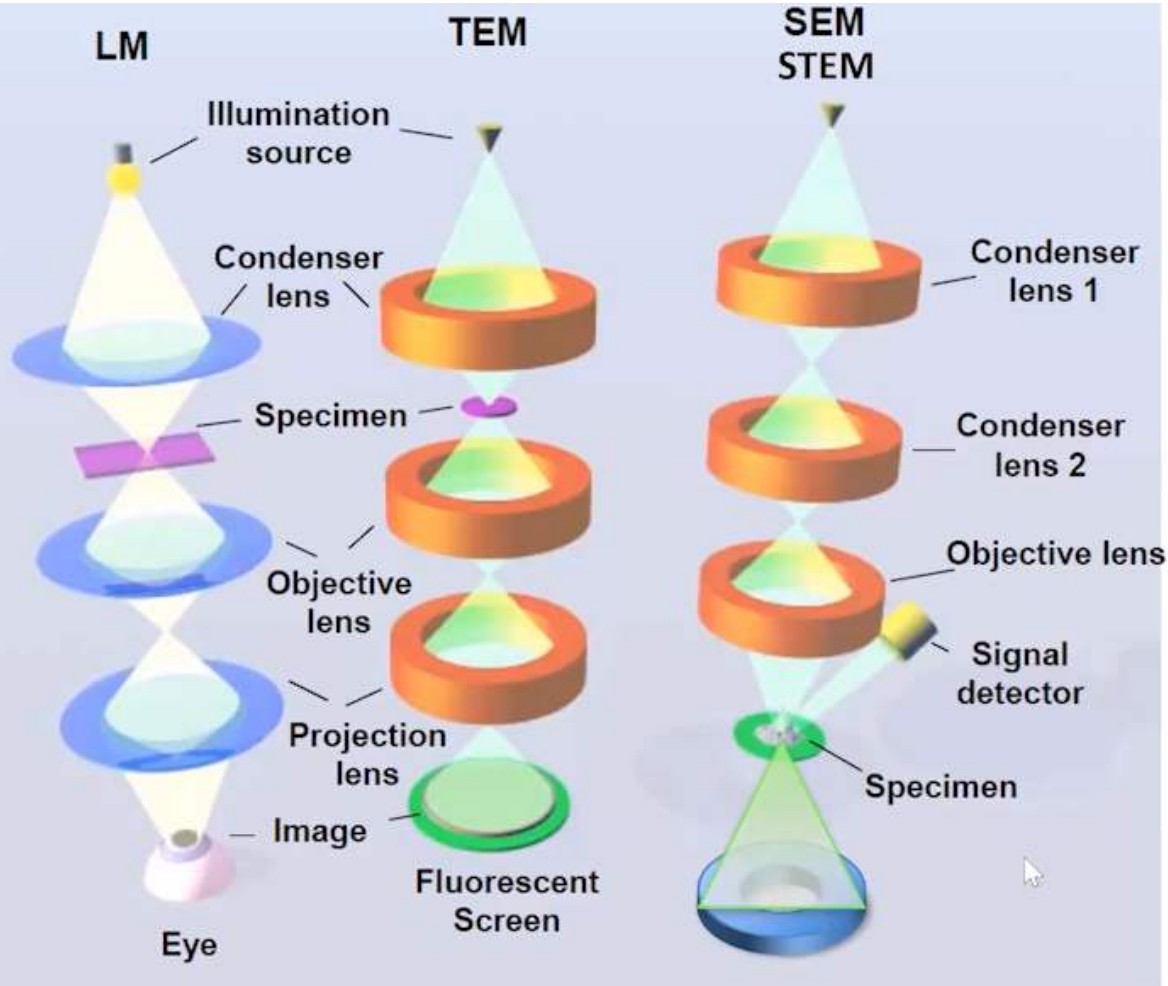
- $p = h/\lambda$ (light also)
- $p = \sqrt{2mE}$
- $E = h^2/2m\lambda^2$

h: planck's constant

$$h = 6.626\,0693(11) \times 10^{-34} \text{ J} \cdot \text{s}$$

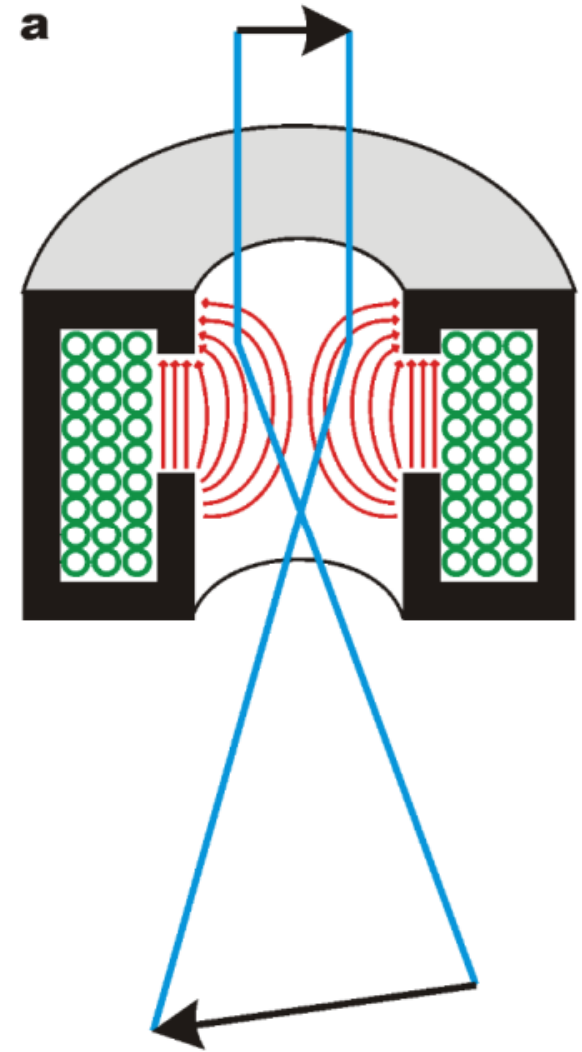
$$h = 4.135\,667\,43(35) \times 10^{-15} \text{ eV} \cdot \text{s}$$

Electron-matter interaction

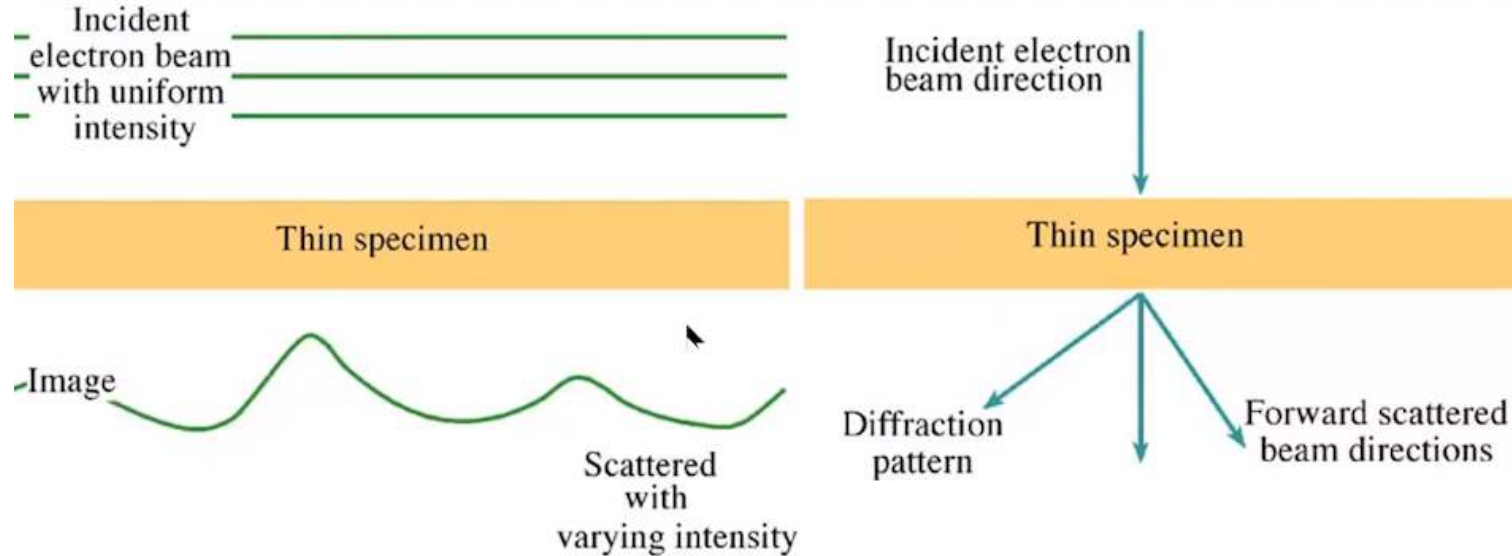


Electron Lenses

- Kawat tembaga (hijau) melingkar di dalam silinder besi (hitam).
- Medan magnet rotasi simetris tetapi radial tidak homogen (merah) memfokuskan berkas elektron (biru).



Scattering in TEM

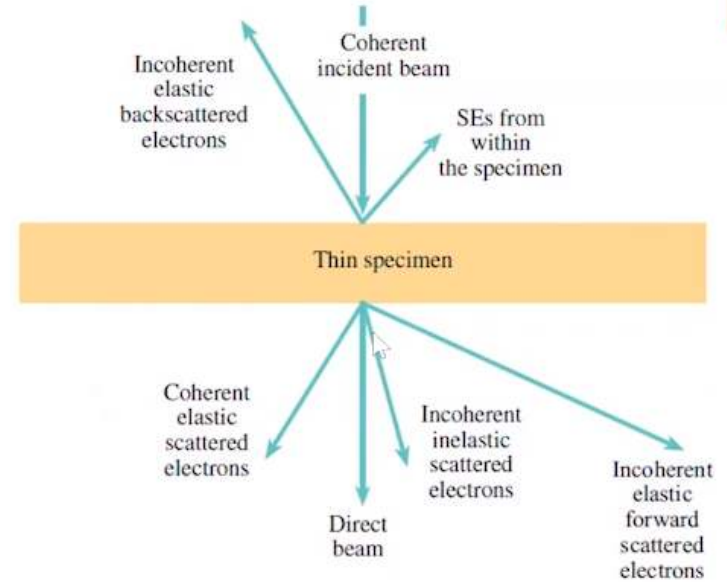


- Scattering in the specimen changes both the spatial and angular distribution of the emerging electrons (exit wave).
- Changes in the spatial distribution (left) are visualized as changes in intensity.
- Changes in the angular distribution are observed as diffraction.

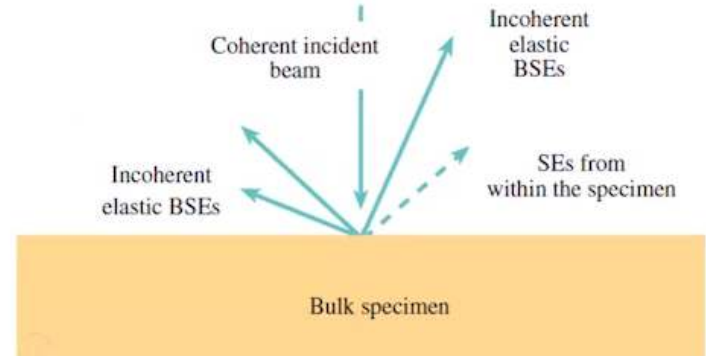
TEM vs SEM

- TEM memiliki forward & back scattering
- SEM hanya back scattering
- Coherent scattering: hubungan kuat antara fase gelombang
- Incoherent: hubungan lemah

TEM

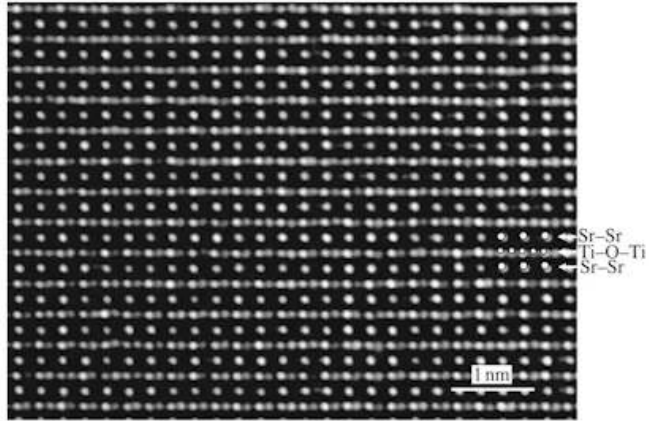


SEM

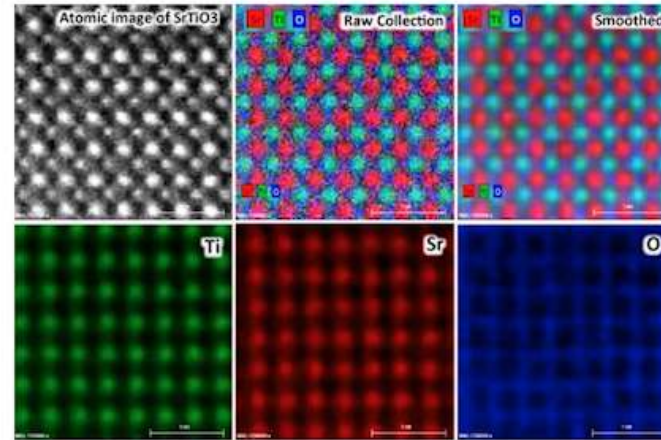


Elastic vs inelastic scattering

K. Urban

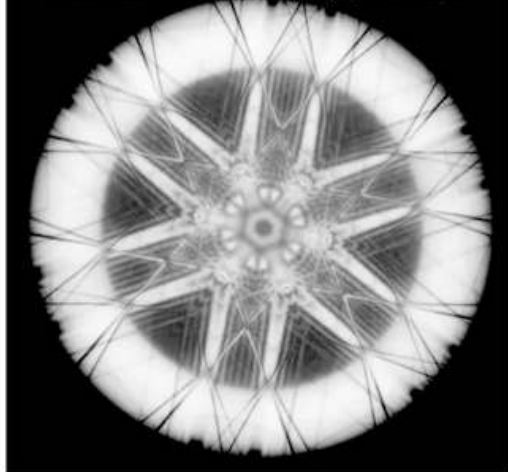


J. LeBeau

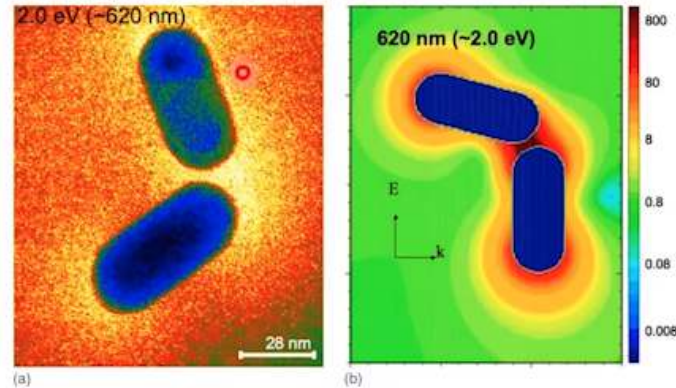


EDS

Hollow-cone LACBED at Si(111) by M.Terauchi

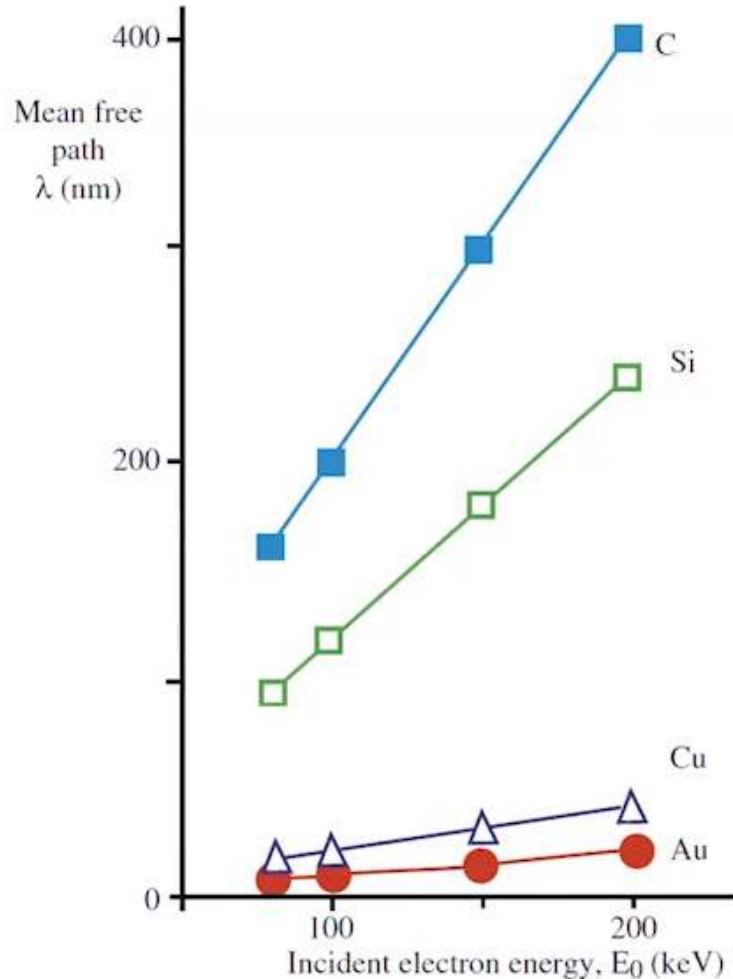


M. N'Gom



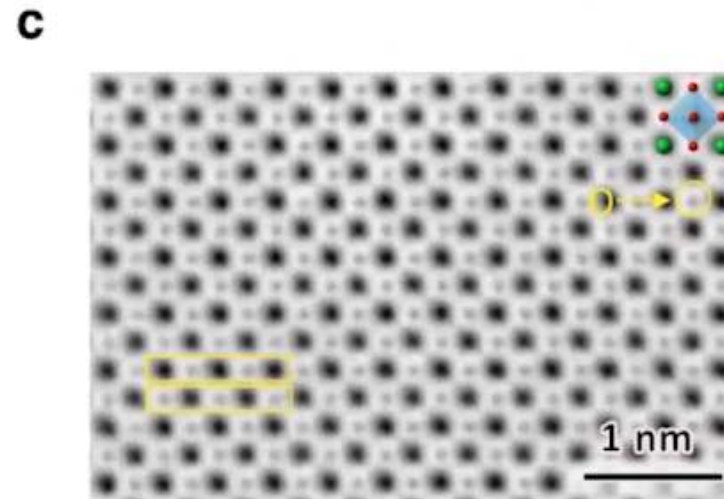
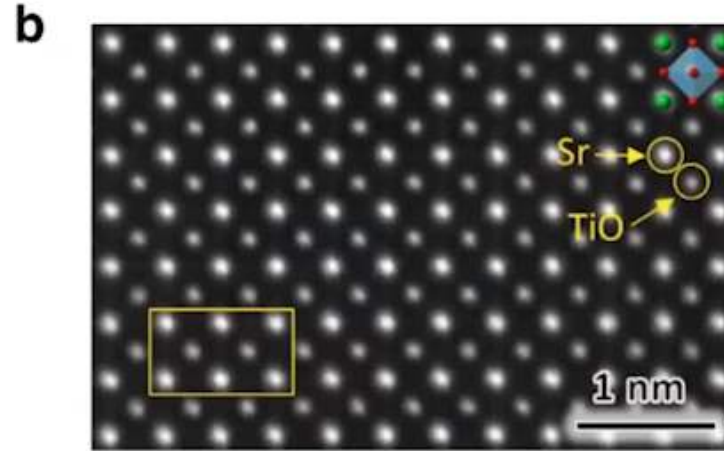
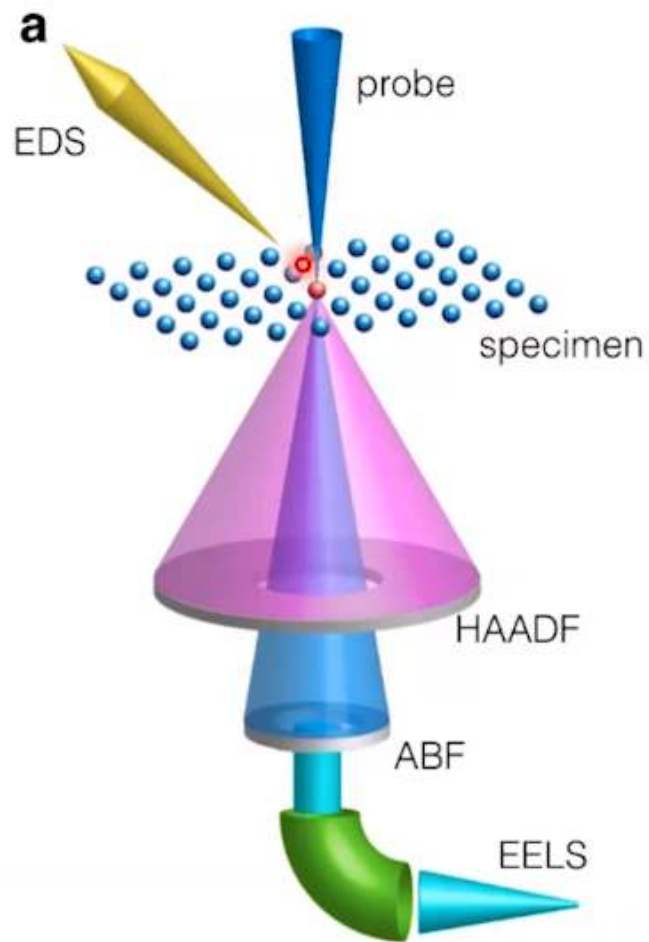
EELS

Sample preparation



- You can reduce the number of statistically likely scattering events by reducing the thickness of your sample below the mean free path of an electron within your material.
- Higher $Z \rightarrow$ thinner sample
- Higher kV $\rightarrow$ thicker sample
- Multiple scattering can lead to uninterpretable images.

Versatility of STEM



Why TEM? HRTEM

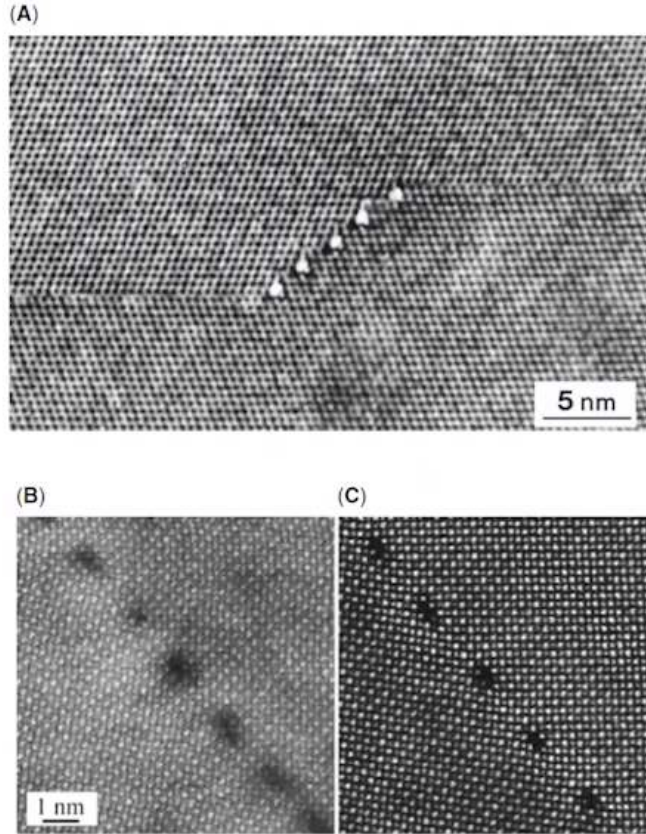
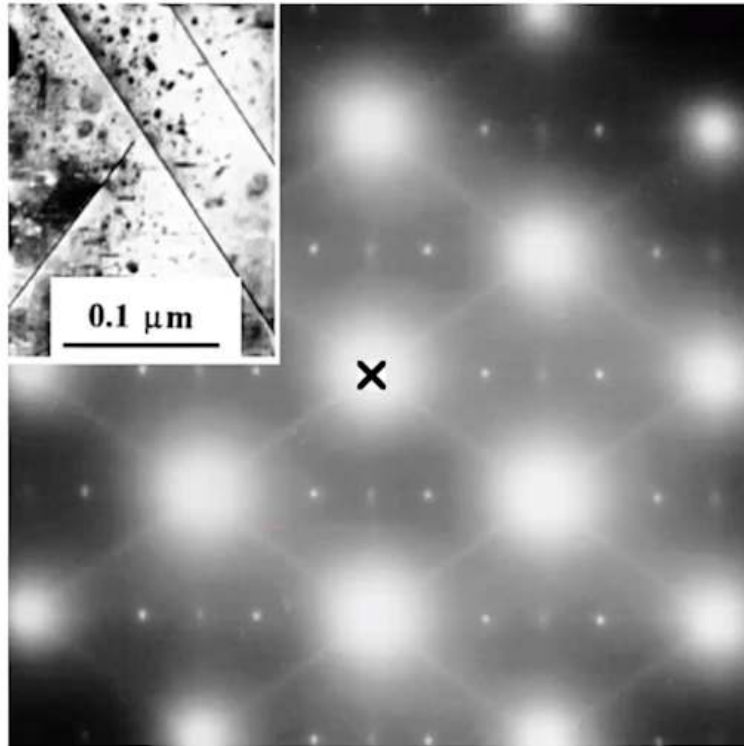


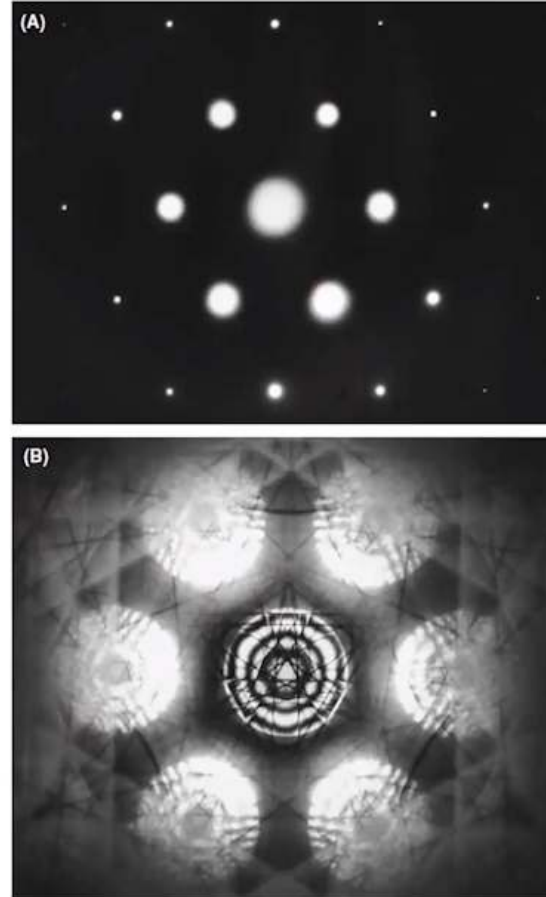
FIGURE 1.2. (A) A twin boundary in spinel stepping from one $\{111\}$ plane to another parallel plane. The white dots are columns of atoms. The change in atomic orientation across the twin boundary can be readily seen even if we do not know what causes the white dots or why indeed they are white. (B) A grain boundary in SrTiO₃ imaged without C_s correction and (C) with C_s correction.

- TEM enables the imaging and study of materials at the atomic length scale!
- As materials scientists, we have a tool which enables the visualization and mapping of the most fundamental length scale of interest!
- Microstructure-property relationships.

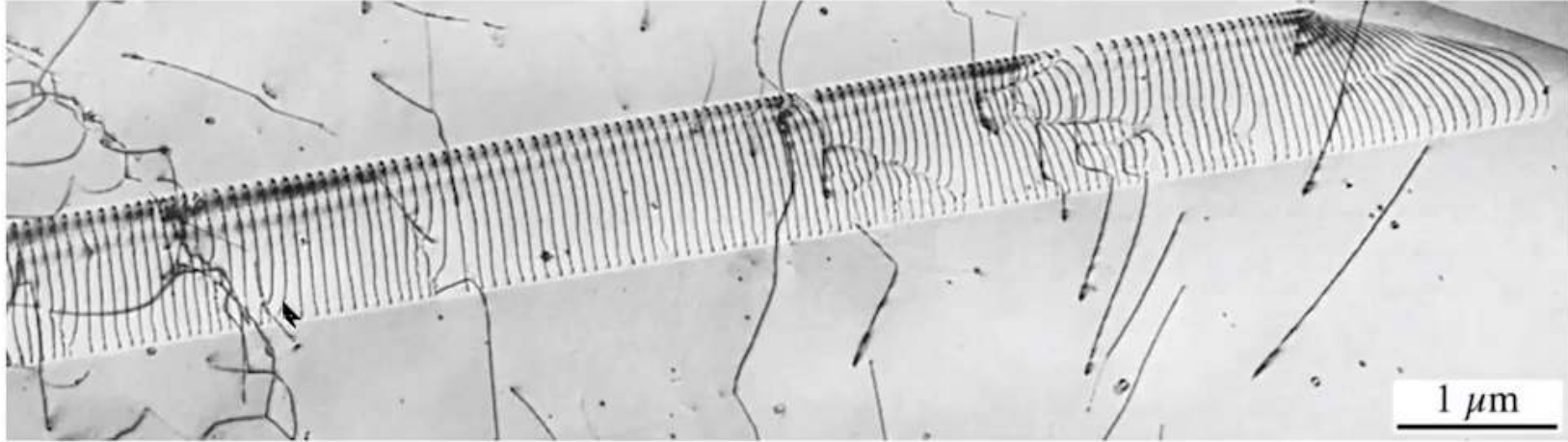
Why TEM? Diffraction



Above: Al-Li-Cu alloy SADP pattern.
Right: (111) Si SADP and CBED patterns



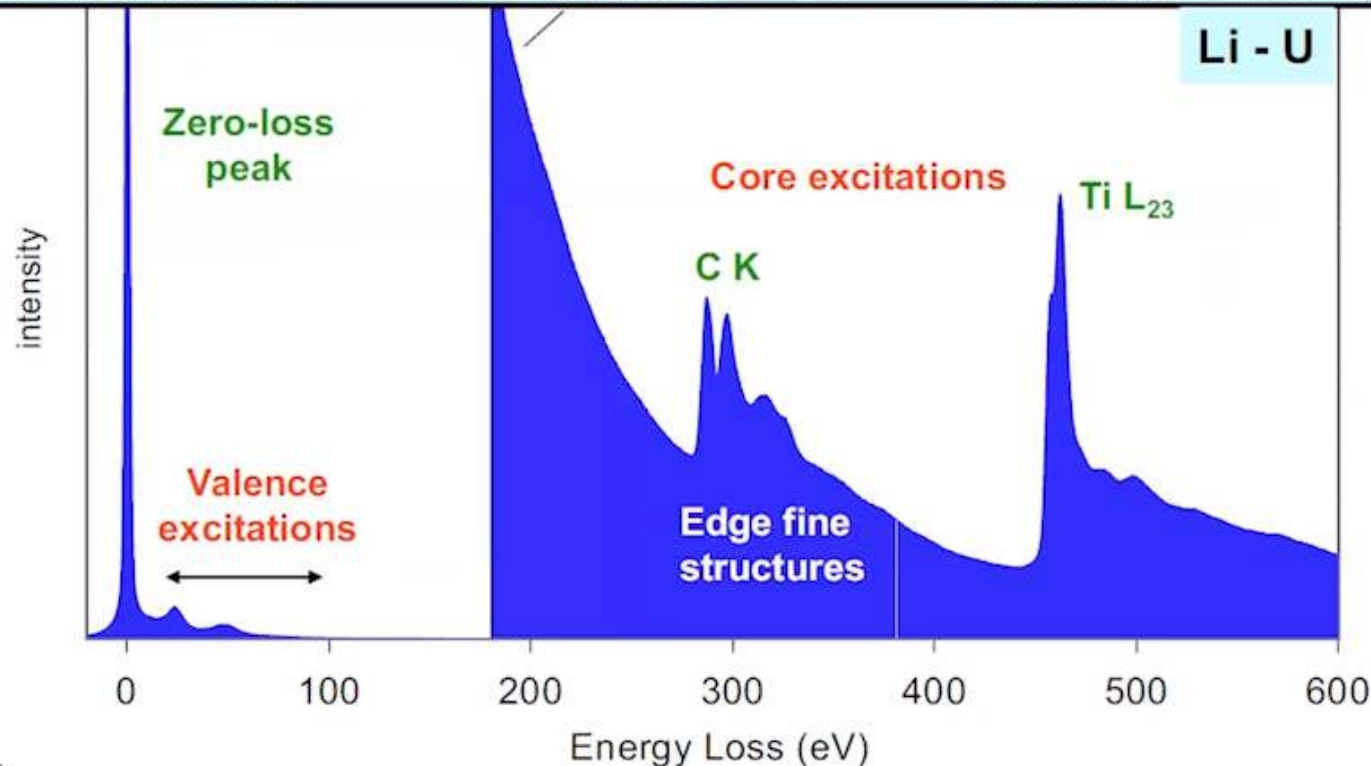
Why TEM? Defect



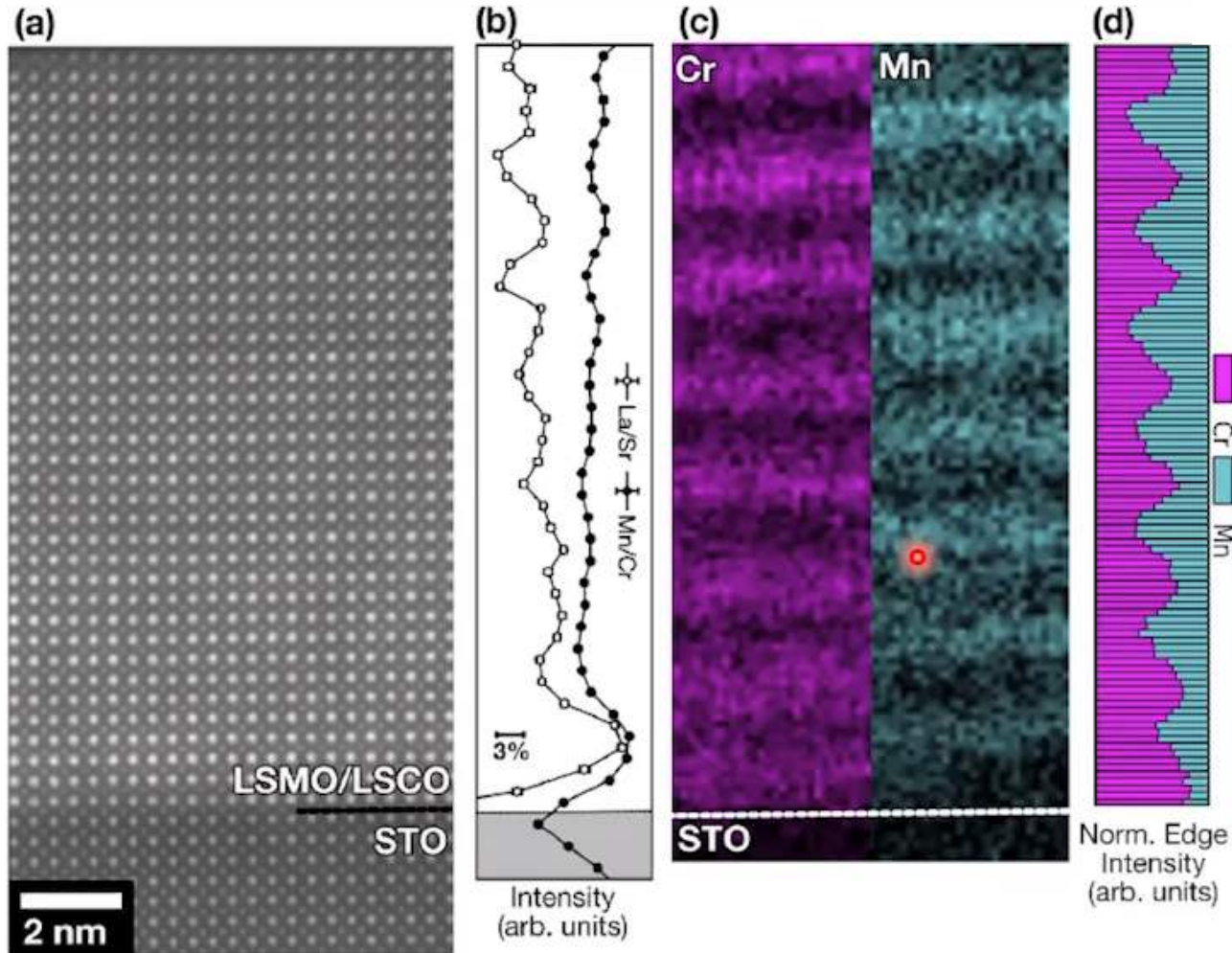
- Because we look inside our material and can image internal structure.
- More importantly, we can image disordering of that internal structure!

Why TEM? EELS as fingerprint

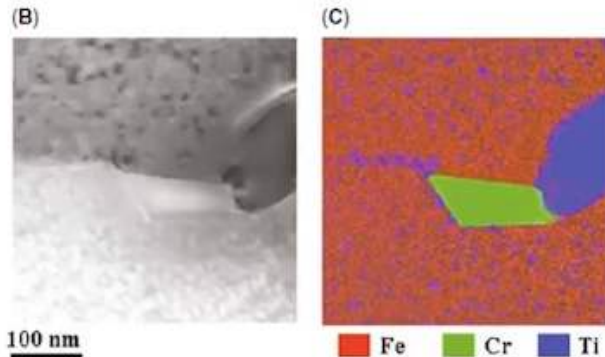
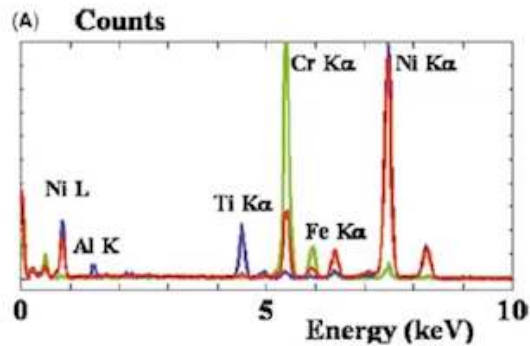
process	energy loss [eV]	θ_E [mrad]	information content
phonons	$\sim 0.02 - 0.10$	5 - 15	"heat" <usually not accessible>
inter-/intra-band transitions	5 - 25	5 - 10	optical properties, band gap
plasmons	$\sim 5 - 25$	< 0.1	free electron density
inner-shell ionizations	$\sim 30 - 1000$ (5000)	1 - 5	element, chemical bonding



Why TEM? EELS as fingerprint



Why TEM? EDS MApping

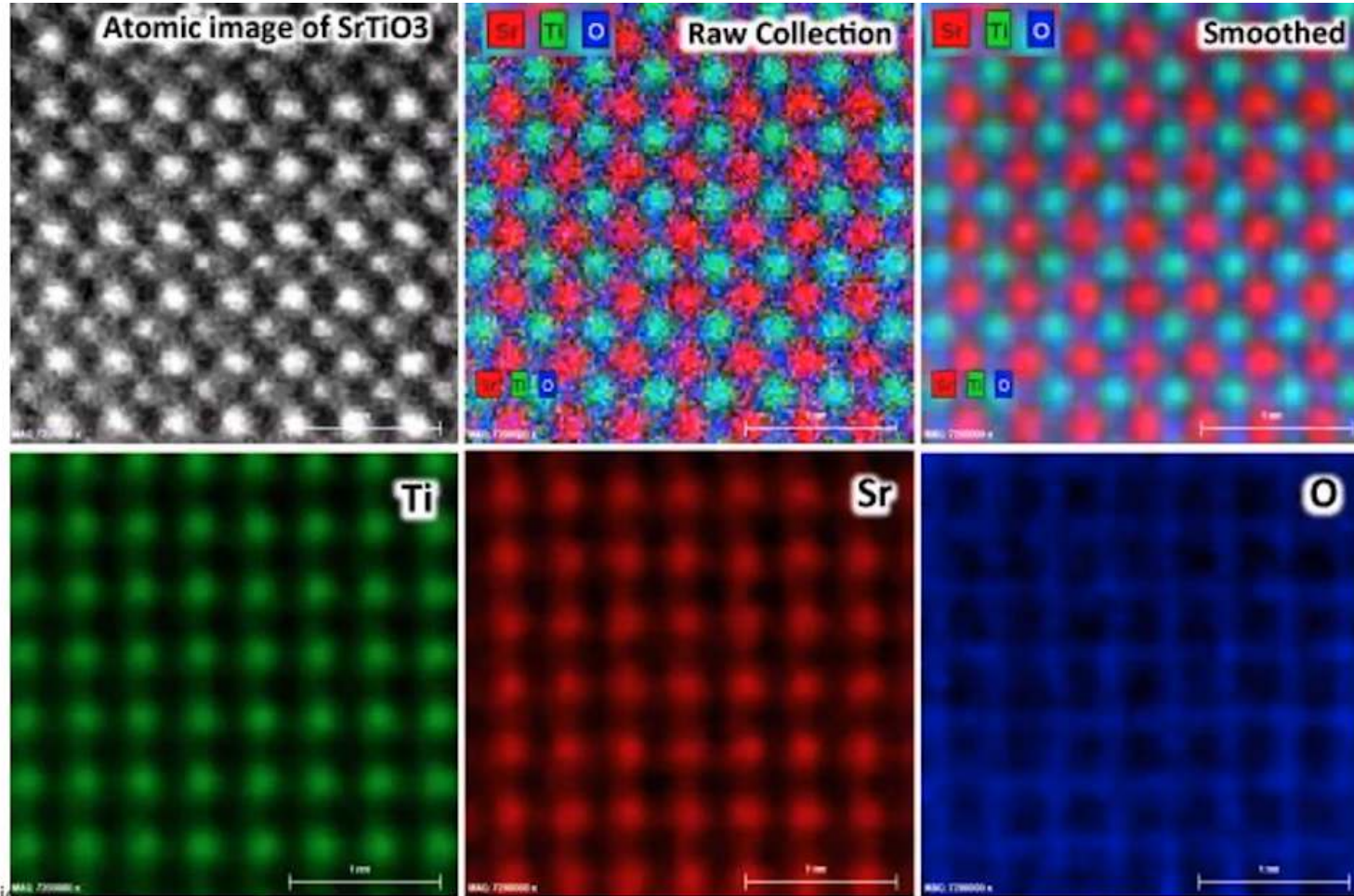


Why STEM? EDS Mapping

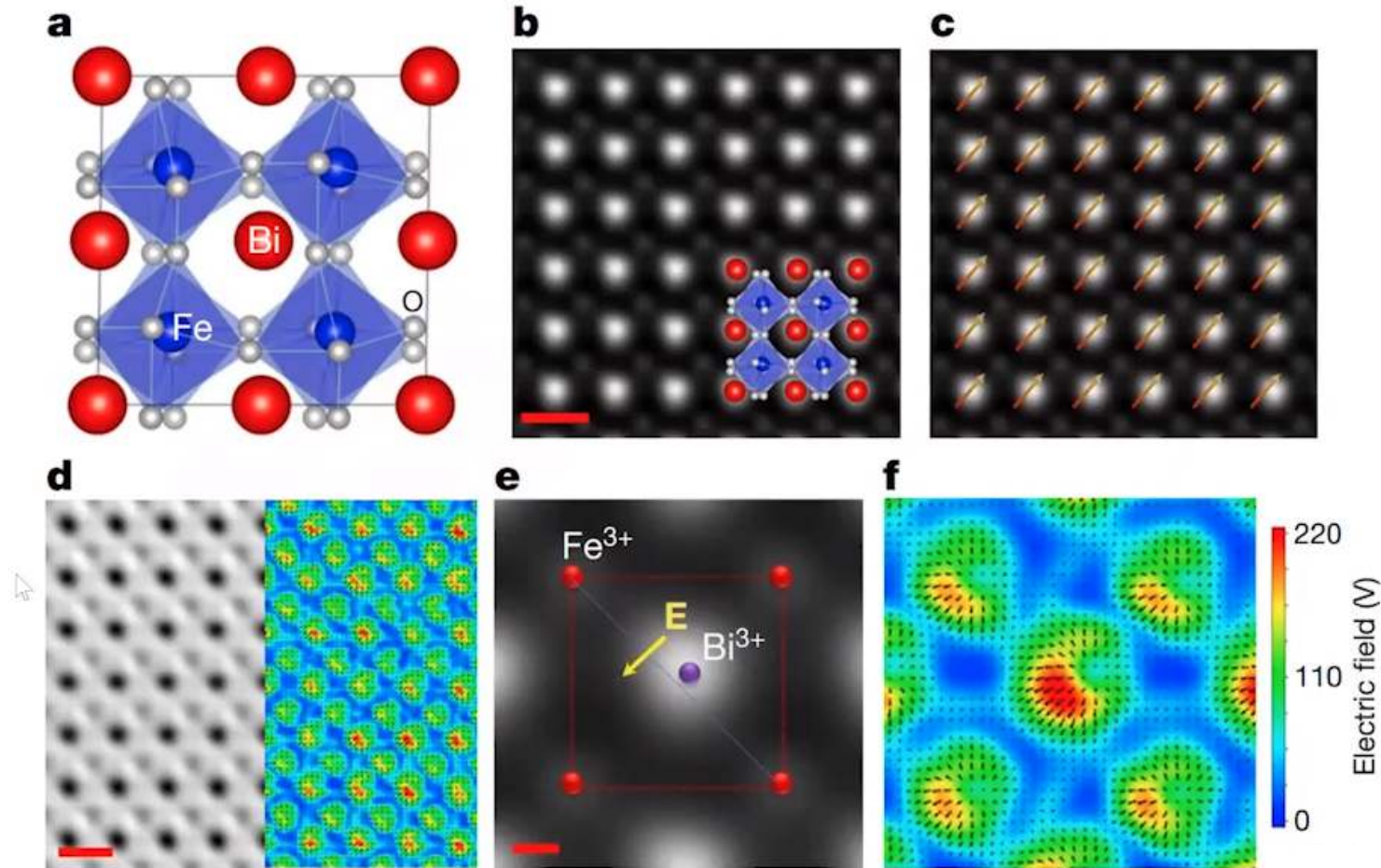
- The high accelerating voltages and bright sources of S/TEMs enables the formation of very small probes with lots of beam current.
- This increases the SNR and spatial resolution of EDS mapping in the STEM.

courtesy: AIF NC State University

Why TEM? EDS MApping



Why STEM? 4D Technique



Ringkasan

- S/TEM has one of the highest spatial resolutions of any imaging technique.
- The spatial resolution allows imaging of atomic structure.
- We can also collect chemical information from individual atoms.
- The image formation physics are complicated!
- S/TEMs are somewhat complicated to operate.
- TEM is 95% sample prep. and 5% microscopy.
- S/TEM is a powerful tool in our materials characterization toolbox.

Tugas

- 1) Pilihlah 1 buah material (**tidak boleh sama dengan yang lain**)
- 2) Carilah hasil citra atau data yang dihasilkan dari hasil interaksi elektron dan bahan (transmitted electron)
- 3) Informasi apa saja yang dapat diperoleh dari citra/data yang anda pilih (baca saja analisa dari pemilik data/paper)

TEM

